

Songhang Li

New York, NY
sl5929@columbia.edu

EDUCATION

Columbia University <i>B.S. in Electrical Engineering</i>	New York, NY Class of 2027
Franklin & Marshall College <i>B.A. in Physics and Mathematics</i> Frank Durrell Enck Memorial Prize in Physics Magna Cum Laude ΦBK	Lancaster, PA Class of 2025
University of California Berkeley <i>Visiting Student, Physics</i>	Berkely, CA Summer 2023
Boston University <i>Visiting Student, Mathematics</i>	Boston, MA Summer 2022

AWARDS & HONORS

The Alma Mater Singer	2025 Commencement, <i>F&M</i>
238th Commencement for Franklin & Marshall College, Songhang Li '25, Baritone. https://www.youtube.com/watch?v=EVJXAYNtoso	
Kershner Scholar in Mathematics	Awarded Spring 2025, <i>F&M</i>
Awarded to students based on their proficiency in Mathematics.	
The Nolt Music Award	Awarded Spring 2025, <i>F&M</i>
Awarded to F&M student musicians to undertake musically enriching projects like senior recital.	
Frank Durrell Enck Memorial Prize in Physics	Awarded Fall 2024, <i>F&M</i>
Established in 1975, traditionally awarded to a graduating Physics major whose overall departmental performance and involvement is most deserving of recognition.	
Summer Scholar Award	Awarded Summer 2024, <i>F&M</i>
Awarded to the student researcher in the summer research with faculties.	
Pi Mu Epsilon Math Honor Society	Since Spring 2024, <i>U.S.A.</i>
Pi Mu Epsilon is a nationally-recognized honorary society dedicated to furthering knowledge of mathematics.	
Sigma Pi Sigma Physic & Astronomy Honor Society	Since Spring 2023, <i>U.S.A.</i>
Sigma Pi Sigma exists to honor outstanding scholarship in physics and astronomy	
Student Independent Research Grants	Awarded Fall 2023, <i>F&M</i>
Awarded to student's advanced research or project whose need exceeds the departmental budgets.	
Kershner Scholar in Physics	Awarded Spring 2023, <i>F&M</i>
Awarded to students based on their proficiency in Physics.	
Dean's and Honors Lists	Awarded Since Fall 2020, <i>F&M</i>
A student earns Honors List recognition for achieving 3.7 or better grade point average.	

PUBLICATION

J.K. Krebs, **Songhang Li**, Dragos Georgescu, and T. D. Pham. (Under review). Eu³⁺ Luminescence as a Probe of Local Crystallinity in Combustion-Synthesized LaAlO₃

Yuxuan Du, Wenya Du, Dabin Lin, Minghao Ai, **Songhang Li**, Lin Zhang. (2023). Recent Progress on Hydrogel-Based Piezoelectric Devices for Biomedical Applications. *Micromachines*, 14(1). doi:10.3390/mi14010167

PRESENTATIONS & CONFERENCES

Mid-Atlantic Section of the American Physical Society Meeting, *Temple University, PA* Nov. 15, 2024

- Rate equation model for fluorescence and phosphorescence in Eu³⁺ doped LaAlO₃.

Student Research Fair, *Franklin and Marshall College, PA* Oct. 18, 2024

- Photon Counting Experiment of Persistent Luminescence in Eu:LAP.

Student Research Fair, *Franklin and Marshall College, PA* Apr. 20, 2024

- Design, construction, testing, and improving: An introduction to Engineering Design Competitions.

ACADEMIC & RESEARCH EXPERIENCES

Photon Counting Experiment of Persistent Luminescence in Eu:LAP (Man. submitted) May 2024 – Jan. 2025
Research Assistant *Advisor: Prof. Ken Krebs, Franklin & Marshall College, Lancaster, PA*

- Measure decay dynamics of excited population of impurity ions after UV laser excitation using photon counting.
- Developed a circuit solution using Arduino to precisely control excitation laser mechanical shutter, enabling photon counting of early state decay.
- Compared the emission spectra measurement with calculated energy levels of Eu³⁺ in LaAlO₃ using density functional theory.
- Conducted a literature review and originated a table summarizing experimental technique and materials.

Society of Automotive Engineers Aero Design Competition Preparation Sept. 2023 – May 2025
Team Leader, Pilot *Advisor: Prof. Ken Krebs, Franklin & Marshall College, Lancaster, PA*

- Designed a power and lift efficient, short-take-off-length 3.8 meters fixed wing plane for water delivery.
- Calculated the lift of the wing based on the structure of air foil in order to improve the load capacity.
- Fabricated the wings utilizing balsa wood and carbon fiber, employing laser cutter and heat-shrink covering film application techniques.
- Led the team members to write the design report and presented the result in research fair.
- Wrote a funding proposal and managed to receive a year-long student research grant from the grant committee.

Automation Algorithm for 4 DOF Robotic Arm Jan. 2023 – May 2023
Researcher *Advisor: Prof. Ken Krebs, Franklin & Marshall College, Lancaster, PA*

- Searched recent publications, textbooks, GitHub robotic arm fabrication and algorithm resources to assist designing a research outline for the group project.
- Designed and built a circuit for touch sensing of a grasping mechanism utilizing piezoresistance sensors.

Electrical and Optical Measurement of Ferroelectrics Material (PLZT) Sept. 2022 – Dec. 2022
Researcher *Advisor: Prof. Ken Krebs, Franklin & Marshall College, Lancaster, PA*

- Investigated the relation between the refraction index and dielectric constants of PLZT film.
- Built multilayer models by ellipsometry data analysis software CompleteEASE to model measurements of the thickness of thin films.
- Measured the refraction index using J.A. Woollam M-2000 spectrum ellipsometry.
- Made a two-plate capacitor with PLZT as dielectric layer to obtain the dielectric constant from capacitance measurements.

- Compared optical and electrical techniques for determination of dielectric constant and refractive index.

Universal Village Program regarding Smart Materials and Intelligent Electronics Jun. 2022 – Aug. 2022
Research Assistant *Advisor: Dr. Lin Zhang, Massachusetts Institute of Technology, Boston, MA*

- Designed a flexible electrode pattern exposure mask using AutoCAD, in preparation for printing of mask.
- Prepared piezoelectric 1-3 composites for piezoelectric sensors.
- Organized the piezoelectric hydrogel sensors comparison table and contributed to manuscript writing, see publication (Micromachines 2023, 14, 167).

SKILLS & INSTRUMENTATION

Hardware & Instrumentation: Oscilloscopes, Waveform Generators, HP LCR, M-2000 J.A. Woollam Spectrum ellipsometry, Monochromator SR430, Stanford Research System Multichannel Photon Counter.

EDA Tools & Design Software: SPICE simulation, MATLAB, Simulink, Arduino IDE, AutoCAD, Autodesk Fusion

Programming language: Python, Java, LaTeX, Mathematica, MATLAB, Arduino, R.

Software: SPICE, IntelliJ, Arduino IDE, AutoCAD, MATLAB, CompleteEASE(optical modeling and ellipsometry data fitting)

Design & Analysis Skills: Analog and digital circuit simulation (AC/DC/transient analysis), Circuit design and debugging, PCB layout and soldering, signal characterization, Data analysis with Python/MATLAB/SciDavis, Characterization of electronic components and semiconductor materials, Embedded prototyping and validation,

Additional Technical Skills: Laser cutting/fabrication (Glowforge), technical documentation (LaTeX, Endnote), design visualization and project planning (X-mind), image processing (Adobe Lightroom, Photoshop)

Language: Mandarin native, Hakka native.

ACTIVITIES

F&M Tennis Club Sept. 2023 – May 2025

- Trained new tennis club members in collaboration with coach

Franklin and Marshall College

Alumni Leaders of Tomorrow, Office of Alumni Engagement

Jun. 2023 – May 2024

- Created alumni connection and networking programs.

Franklin and Marshall College

Vocal

Sept. 2021 – May 2025

Recital and College Chorus singer

Franklin and Marshall College

- Sang Alma Mater at Class of 2025 Commencement
- Participated in College Chorus and performed at important school events like True Blue Weekend.
- Sang in vocal lesson recital at the end of every semester.

EXPERIENCES

Department of Physics and Astronomy

Jan. 2024 – May 2025

Teaching and Research Assistant

Franklin and Marshall College, Lancaster, PA

- Compared and contrasted two different textbooks and completed textbook exercises to help the professor mark the key points of instruction for PHY321 Introduction to Electronics.
- Experimented with different teaching kits chosen by the professor to facilitate in selecting a kit that is more conducive to teaching and learning.

Career Center

Sept. 2022 – May 2025

Photographer

Franklin and Marshall College, Lancaster, PA

- Took LinkedIn headshots and event pictures at career service center.

INTERESTS

RC airplane, singing, photography (fashion and documentary, film, and darkroom processing), tennis, swimming, cycling.